

5G RAN

Interference Avoidance Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....

2 About This Document.....

2.1 General Statements.....

2.2 Features in This Document.....

2.3 Differences.....

3 Overview.....

4 Interference Between Intra-Frequency Neighboring NR Cells.....

4.1 Time-Domain Interference Avoidance (Low-Frequency TDD).....

4.1.1 Principles.....

4.1.2 Network Analysis.....

4.1.2.1 Benefits.....

4.1.2.2 Impacts.....

4.1.3 Requirements.....

4.1.3.1 Licenses.....

4.1.3.2 Functions.....

4.1.3.2.1 Prerequisite Functions.....

4.1.3.2.2 Mutually Exclusive Functions.....

4.1.3.2.3 Function Impacts.....

4.1.3.3 Hardware.....

4.1.3.4 Cells.....

4.1.3.5 Others.....

4.1.4 Operation and Maintenance.....

4.1.4.1 Data Configuration.....

4.1.4.1.1 Data Preparation.....

4.1.4.1.2 Using MML Commands.....

4.1.4.1.3 Using the MAE-Deployment.....

4.1.4.2 Activation Verification.....

4.1.4.3 Network Monitoring.....

4.2 Common Channel Interference Avoidance.....

4.2.1 Principles.....

4.2.2 Network Analysis.....

4.2.2.1 Benefits.....

1

3

3

4

4

6

9

9

9

11

12

12

12

12

12

12

12

13

14

14

15

15

15

15

17

18

18

18

18

18

19

19

4.2.2.2 Impacts.....	20
4.2.3 Requirements.....	20
4.2.3.1 Licenses.....	20
4.2.3.2 Functions.....	21
4.2.3.2.1 Prerequisite Functions.....	21
4.2.3.2.2 Mutually Exclusive Functions.....	21
4.2.3.2.3 Function Impacts.....	21
4.2.3.3 Hardware.....	23
4.2.3.4 Others.....	23
4.2.4 Operation and Maintenance.....	23
4.2.4.1 Data Configuration.....	23
4.2.4.1.1 Data Preparation.....	23
4.2.4.1.2 Using MML Commands.....	24
4.2.4.1.3 Using the MAE-Deployment.....	24
4.2.4.2 Activation Verification.....	25
4.2.4.3 Network Monitoring.....	25
5 Intra-Frequency Interference in LTE and NR Co-Coverage Scenarios.....	26
5.1 Interference-based Subband Robust Weights (Low-Frequency TDD).....	26
5.1.1 Principles.....	26
5.1.2 Network Analysis.....	29
5.1.2.1 Benefits.....	29
5.1.2.2 Impacts.....	29
5.1.3 Requirements.....	29
5.1.3.1 Licenses.....	30
5.1.3.2 Functions.....	30
5.1.3.2.1 Prerequisite Functions.....	30
5.1.3.2.2 Mutually Exclusive Functions.....	31
5.1.3.2.3 Function Impacts.....	32
5.1.3.3 Hardware.....	33
5.1.3.4 Others.....	34
5.1.4 Operation and Maintenance.....	34
5.1.4.1 Data Configuration.....	34
5.1.4.1.1 Data Preparation.....	34
5.1.4.1.2 Using MML Commands.....	36
5.1.4.1.3 Using the MAE-Deployment.....	36
5.1.4.2 Activation Verification.....	37
5.1.4.3 Network Monitoring.....	37
6 Common Interference Avoidance Technologies.....	38
6.1 Frequency Offset Measurement Optimization Based on Interference Suppression (Low-Frequency TDD).....	38
6.1.1 Principles.....	38
6.1.2 Network Analysis.....	39

6.1.2.1 Benefits..... 39

6.1.2.2 Impacts..... 39

6.1.3 Requirements..... 39

6.1.3.1 Licenses..... 39

6.1.3.2 Functions..... 39

6.1.3.2.1 Prerequisite Functions..... 40

6.1.3.2.2 Mutually Exclusive Functions..... 40

6.1.3.2.3 Function Impacts..... 40

6.1.3.3 Hardware..... 41

6.1.3.4 Cells..... 41

6.1.4 Operation and Maintenance..... 41

6.1.4.1 Data Configuration..... 41

6.1.4.1.1 Data Preparation..... 41

6.1.4.1.2 Using MML Commands..... 41

6.1.4.1.3 Using the MAE-Deployment..... 42

6.1.4.2 Activation Verification..... 42

6.1.4.3 Network Monitoring..... 42

6.2 External Interference Avoidance (Low-Frequency TDD)..... 42

6.2.1 Principles..... 42

6.2.2 Network Analysis..... 48

6.2.2.1 Benefits..... 48

6.2.2.2 Impacts..... 48

6.2.3 Requirements..... 48

6.2.3.1 Licenses..... 48

6.2.3.2 Functions..... 49

6.2.3.2.1 Prerequisite Functions..... 49

6.2.3.2.2 Mutually Exclusive Functions..... 50

6.2.3.2.3 Function Impacts..... 56

6.2.3.3 Hardware..... 58

6.2.3.4 Cells..... 59

6.2.3.5 Others..... 59

6.2.4 Operation and Maintenance..... 59

6.2.4.1 Data Configuration..... 59

6.2.4.1.1 Data Preparation..... 59

6.2.4.1.2 Using MML Commands..... 61

6.2.4.1.3 Using the MAE-Deployment..... 62

6.2.4.2 Activation Verification..... 63

6.2.4.3 Network Monitoring..... 63

6.3 Cell-level PUSCH Interference Avoidance (Low-Frequency TDD)..... 63

6.3.1 Principles..... 63

6.3.2 Network Analysis..... 66

6.3.2.1 Benefits..... 67

6.3.2.2 Impacts..... 67

6.3.3 Requirements..... 67

6.3.3.1 Licenses.....67

6.3.3.2 Functions..... 67

6.3.3.2.1 Prerequisite Functions..... 67

6.3.3.2.2 Mutually Exclusive Functions.....68

6.3.3.2.3 Function Impacts..... 69

6.3.3.3 Hardware..... 69

6.3.3.4 Cells.....70

6.3.3.5 Others..... 70

6.3.4 Operation and Maintenance..... 70

6.3.4.1 Data Configuration.....70

6.3.4.1.1 Data Preparation.....70

6.3.4.1.2 Using MML Commands..... 72

6.3.4.1.3 Using the MAE-Deployment..... 73

6.3.4.2 Activation Verification..... 73

6.3.4.3 Network Monitoring..... 73

6.4 GP Symbol Shutdown in Special Slots..... 74

6.4.1 Principles..... 74

6.4.2 Network Analysis..... 74

6.4.2.1 Benefits..... 74

6.4.2.2 Impacts..... 75

6.4.3 Requirements..... 75

6.4.3.1 Licenses.....75

6.4.3.2 Functions.....75

6.4.3.2.1 Prerequisite Functions..... 75

6.4.3.2.2 Mutually Exclusive Functions.....75

6.4.3.2.3 Function Impacts..... 75

6.4.3.3 Hardware..... 76

6.4.3.4 Cells.....76

6.4.3.5 Others..... 76

6.4.4 Operation and Maintenance..... 76

6.4.4.1 Data Configuration.....76

6.4.4.1.1 Data Preparation.....76

6.4.4.1.2 Using MML Commands..... 77

6.4.4.1.3 Using the MAE-Deployment..... 77

6.4.4.2 Activation Verification..... 77

6.4.4.3 Network Monitoring..... 78

7 Parameters..... 79

8 Counters..... 81

9 Glossary..... 82

10 Reference Documents..... 83

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added GP symbol shutdown in special slots. For details, see 6.4 GP Symbol Shutdown in Special Slots .	Modified parameter: Added the SS_GAP_SYMBOL_SHUTDN_SW option to the NRDUCellTimeIntrf.TimeIntrfAvoidSw parameter.	TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between self-fusion weights and interference-based subband robust weights. For details, see 5.1.3.2.3 Function Impacts .	None	TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between cell-level PUSCH interference avoidance and CORESET 0 offset policy configuration. For details, see 6.3.3.2.2 Mutually Exclusive Functions .	None	TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between frequency offset measurement optimization based on interference suppression and interference-based precise frequency offset estimation. For details, see 6.1.3.2.3 Function Impacts .	None	TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
TD D	FOFD-081208	Inter-Mode Interference Suppression	5.1 Interference-based Subband Robust Weights (Low-Frequency TDD)

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Time-domain interference avoidance (low-frequency TDD)	None	4.1 Time-Domain Interference Avoidance (Low-Frequency TDD)
Common channel interference avoidance	None	4.2 Common Channel Interference Avoidance
Interference-based subband robust weights (low-frequency TDD)	None	5.1 Interference-based Subband Robust Weights (Low-Frequency TDD)

Function Name	Difference	Chapter/Section
Frequency offset measurement optimization based on interference suppression (low-frequency TDD)	None	6.1 Frequency Offset Measurement Optimization Based on Interference Suppression (Low-Frequency TDD)
External interference avoidance (low-frequency TDD)	None	6.2 External Interference Avoidance (Low-Frequency TDD)
Cell-level PUSCH interference avoidance (low-frequency TDD)	None	6.3 Cell-level PUSCH Interference Avoidance (Low-Frequency TDD)

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Time-domain interference avoidance (low-frequency TDD)	Supported only in low frequency bands	4.1 Time-Domain Interference Avoidance (Low-Frequency TDD)
Common channel interference avoidance	Supported only in low frequency bands	4.2 Common Channel Interference Avoidance
Interference-based subband robust weights (low-frequency TDD)	Supported only in low frequency bands	5.1 Interference-based Subband Robust Weights (Low-Frequency TDD)
Frequency offset measurement optimization based on interference suppression (low-frequency TDD)	Supported only in low frequency bands	6.1 Frequency Offset Measurement Optimization Based on Interference Suppression (Low-Frequency TDD)
External interference avoidance (low-frequency TDD)	Supported only in low frequency bands	6.2 External Interference Avoidance (Low-Frequency TDD)
Cell-level PUSCH interference avoidance (low-frequency TDD)	Supported only in low frequency bands	6.3 Cell-level PUSCH Interference Avoidance (Low-Frequency TDD)

3 Overview

With the rapid growth of the wireless communications market—especially the explosive increase in data services in recent years—requirements for network coverage and capacity have become increasingly demanding. To meet these requirements, operators have invested heavily in large-scale network deployments across multiple RATs. As cell radii shrink and the network noise floor continues to rise, interference scenarios have become more complex, significantly degrading user experience. To mitigate the impact of different types of interference on networks, various interference avoidance functions are introduced. [Table 3-1](#) lists different types of interference, while [Table 3-2](#) and [Table 3-3](#) provide mappings of uplink and downlink interference avoidance functions, respectively. In this context, uplink interference refers to interference affecting a base station's uplink channels, whereas downlink interference refers to interference affecting a base station's downlink channels.

Table 3-1 Interference

Interference	Description
Interference between intra-frequency neighboring NR cells	This type of interference refers to intra-frequency interference between intra-frequency neighboring NR cells in NR intra-frequency networking scenarios.
Intra-frequency interference in LTE and NR co-coverage scenarios	In LTE and NR co-coverage scenarios with the spectrum of an LTE cell overlapping that of an NR cell, the LTE cell causes intra-frequency interference to the NR cell. Such interference is referred to as intra-frequency interference in LTE and NR co-coverage scenarios.

Table 3-2 Mapping for uplink interference avoidance functions

Interference	Feature/ Function Name	Principle	Chapter/Section
Interference between intra-frequency neighboring NR cells	Time-domain interference avoidance (low-frequency TDD)	Interference avoidance is performed in the time domain of a victim cell.	4.1 Time-Domain Interference Avoidance (Low-Frequency TDD)
Intra-frequency interference in LTE and NR co-coverage scenarios	Interference-based subband robust weights (low-frequency TDD)	A subband robust weight is calculated based on SRS measurements over a non-interference range and is used across the entire band.	5.1 Interference-based Subband Robust Weights (Low-Frequency TDD)
Common interference	Frequency offset measurement optimization based on interference suppression (low-frequency TDD)	Interference suppression is performed during frequency offset estimation to change the directions of uplink receive beams, thereby avoiding interference.	6.1 Frequency Offset Measurement Optimization Based on Interference Suppression (Low-Frequency TDD)
Common interference	External interference avoidance (low-frequency TDD)	A range within which interference is present (an interference-present range or an interference range) is manually configured or automatically identified, and the base station determines whether this range can be used for uplink and downlink scheduling based on the measured interference strength.	6.2 External Interference Avoidance (Low-Frequency TDD)

Interference	Feature/ Function Name	Principle	Chapter/Section
Common interference	Cell-level PUSCH interference avoidance (low-frequency TDD)	A range within which interference is present is manually configured, and the base station determines whether this range can be used for uplink scheduling based on the measured interference strength.	6.3 Cell-level PUSCH Interference Avoidance (Low-Frequency TDD)
Common interference	GP symbol shutdown in special slots	The gNodeB shuts down the first two GP symbols of a special slot and postpones the timing of downlink-to-uplink switching, thereby reducing the impact of interference in the GP symbols on uplink performance.	6.4 GP Symbol Shutdown in Special Slots

Table 3-3 Mapping for downlink interference avoidance functions

Interference	Feature/ Function Name	Principle	Chapter/Section
Interference between intra-frequency neighboring NR cells	Common channel interference avoidance	The base station avoids downlink scheduling in a cell at the time-frequency positions used by neighboring-cell common channels if it determines that the cell is affected by interference from those channels.	4.2 Common Channel Interference Avoidance

 NOTE

The **N_UL_NI_Avg** counter can be used to observe the average interference and noise received on each uplink PRB. If the interference and noise power is imbalanced in the time domain within a sampling period of 1s, the value of this counter will increase or decrease after cell reestablishment.

4 Interference Between Intra-Frequency Neighboring NR Cells

Interference between intra-frequency neighboring NR cells refers to intra-frequency interference between intra-frequency neighboring NR cells in NR intra-frequency networking scenarios.

4.1 Time-Domain Interference Avoidance (Low-Frequency TDD)

4.1.1 Principles

In NR intra-frequency networking scenarios, interference exists between cells that provide continuous coverage and are configured with different self-contained slot structures. The interference principles are as follows.

Figure 4-1 Time-domain interference between cells with different self-contained slot structures

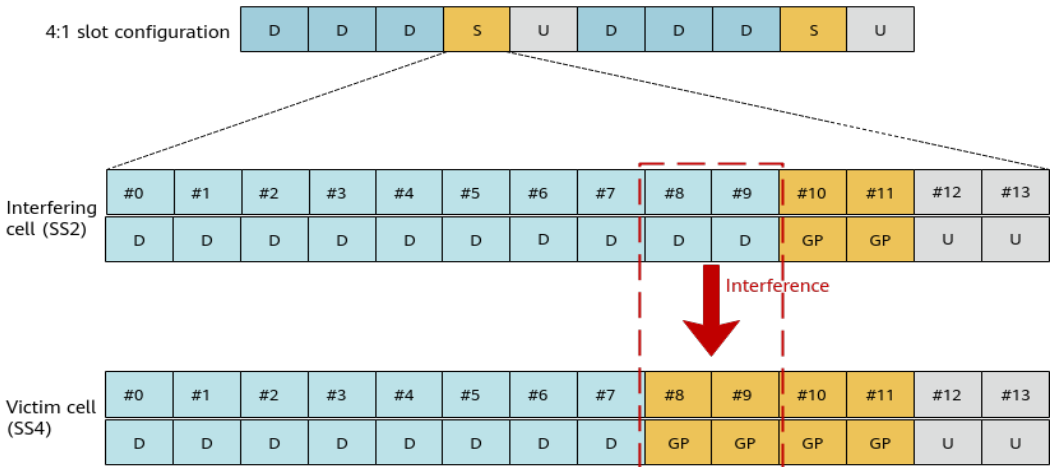


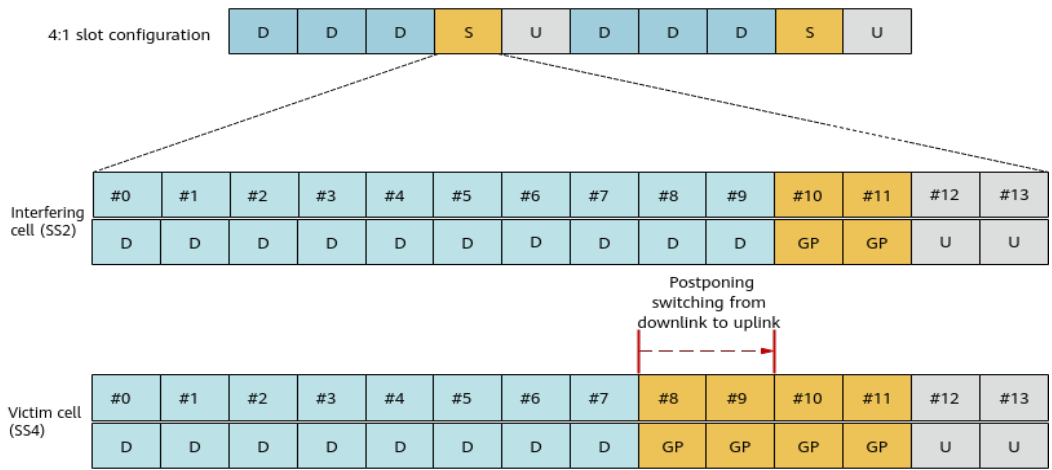
Figure 4-1 shows an example wherein the self-contained slot structures of two cells are separately SS2 and SS4, with the 4:1 (DDDSU) slot configuration used. In

this example, the timing of switching from downlink to uplink differs between the two cells due to the use of different self-contained slot structures. This is because switching from downlink to uplink occurs upon the completion of transmission in the last downlink symbol of a self-contained slot. To illustrate, the interfering cell switches from a downlink transmission state to an uplink reception state upon the completion of data transmission in symbol 9, while such switching occurs in the victim cell upon the completion of data transmission in symbol 7.

In NR intra-frequency continuous networking scenarios, when the coverage areas of the two cells overlap, the downlink signals transmitted in symbols 8 and 9 of the interfering cell will interfere with the uplink reception in symbols 8 and 9 of the victim cell. Consequently, channel calibration may fail in the victim cell, leading to performance deterioration. If the instantaneous strength of interference is high, the performance deterioration will be sharp.

Against this backdrop, the time-domain interference avoidance function is introduced. **This function postpones switching from downlink to uplink in victim cells to realize interference avoidance in the time domain**, as shown in [Figure 4-2](#).

Figure 4-2 Time-domain interference avoidance



This function works in multiple self-contained slot structure combination scenarios, and its function switch (an option of the **gNodeBAlgo.TimeIntrfAvoidSw** parameter) varies in these scenarios. For details, see [Table 4-1](#).

Table 4-1 Mapping between self-contained slot structure combinations and options of **gNodeBAlgo.TimeIntrfAvoidSw**

Self-Contained Slot Structure of the Victim Cell	Self-Contained Slot Structure of the Interfering Cell	Option of the gNodeBAlgo.TimeIntrfAvoidSw Parameter
SS56	SS54	SS56_4GP_TIME_INTRF_AVOID_SW
SS518	SS54	SS518_4GP_TIME_INTRF_AVOID_SW

Self-Contained Slot Structure of the Victim Cell	Self-Contained Slot Structure of the Interfering Cell	Option of the gNodeBAlgo.TimeIntrfAvoidSw Parameter
SS86	SS84	SS86_4GP_TIME_INTRF_AVOID_SW
SS818	SS84	SS818_4GP_TIME_INTRF_AVOID_SW
SS4	SS2	SS4_2GP_TIME_INTRF_AVOID_SW
SS6	SS2	SS6_2GP_TIME_INTRF_AVOID_SW
SS6	SS4	SS6_4GP_TIME_INTRF_AVOID_SW
SS01001202	SS2	SS01001202_2GP_INTRF_AVOID_SW
SS01001202	SS4	SS01001202_4GP_INTRF_AVOID_SW
SS18	SS2	SS18_2GP_TIME_INTRF_AVOID_SW
SS18	SS4	SS18_4GP_TIME_INTRF_AVOID_SW
SS54	SS52	SS54_2GP_TIME_INTRF_AVOID_SW
SS54	SS51	SS54_1GP_TIME_INTRF_AVOID_SW

NOTE

The requirements for selecting the options of the **gNodeBAlgo.TimeIntrfAvoidSw** parameter are as follows:

- If an option needs to be selected for a victim base station, the self-contained slot structure of at least one cell served by the base station must be the same as the "Self-Contained Slot Structure of the Victim Cell" value corresponding to this option in the preceding table. For example, if the **SS56_4GP_TIME_INTRF_AVOID_SW** option of the **gNodeBAlgo.TimeIntrfAvoidSw** parameter needs to be selected for a victim base station, the self-contained slot structure of at least one cell served by the base station must be SS56.
- Different options cannot be selected at the same time if these options correspond to the same "Self-Contained Slot Structure of the Victim Cell" value. For example, the **SS18_2GP_TIME_INTRF_AVOID_SW** and **SS18_4GP_TIME_INTRF_AVOID_SW** options of the **gNodeBAlgo.TimeIntrfAvoidSw** parameter cannot be selected at the same time.

The procedure of enabling time-domain interference avoidance is as follows:

1. Select a victim base station. In NR intra-frequency continuous networking scenarios, if the self-contained slot structures of cells under different base stations are different, the base station with a larger number of GP symbols is the victim base station.
2. Select the required options of the **gNodeBAlgo.TimeIntrfAvoidSw** parameter for the victim base station based on the self-contained slot structures of the cells under the interfering and victim base stations. For details, see [Table 4-1](#).

4.1.2 Network Analysis

4.1.2.1 Benefits

This function mitigates time-domain interference between base stations with different self-contained slot structures in intra-frequency continuous networking scenarios. This function is recommended if adjacent base stations adopt different self-contained slot structures. After this function is enabled, the value of **N.UL.NI.Avg** decreases and the **User Uplink Average Throughput (DU)** increases for victim cells.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

Time-domain interference avoidance impacts specific indicators, as described below:

- When the number of GP symbols for time-domain interference avoidance is 4 for the victim base station, the value of **N.GAP.FifthSymbolFromLast.Pwr** is invalid.
- When the number of GP symbols for time-domain interference avoidance is 2 for the victim base station, the values of **N.GAP.ThirdSymbolFromLast.Pwr** and **N.GAP.FifthSymbolFromLast.Pwr** are invalid.
- When the number of GP symbols for time-domain interference avoidance is 1 for the victim base station, the value of **N.GAP.ThirdSymbolFromLast.Pwr** is invalid.

4.1.3 Requirements

4.1.3.1 Licenses

None

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

None

4.1.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When the cell self-contained slot structure is SS54, the SS54_1GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimeIntrfAvoidSw parameter and the INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter cannot be both selected.
Inter-RRU channel calibration for distributed massive MIMO	DM_MIMO_INTER_RRU_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	When the cell self-contained slot structure is SS54, the SS54_1GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimeIntrfAvoidSw parameter and the DM_MIMO_INTER_RRU_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter cannot be both selected.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	When the self-contained slot structure of a cell is SS54 and the NRDUCell.NrDuCellNetworkingMode parameter is set to FUSION_CELL , the SS54_1GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimeIntrfAvoidSw parameter cannot be selected.
Self-contained slot scheduling optimization	gNodeBAlgo.SSlotSchOpt set to UL_SCHEDULE_OPT or DL_UL_SCHEDULE_OPT	None	When the SS54_2GP_TIME_INTRF_AVOID_SW or SS54_1GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimeIntrfAvoidSw parameter is selected, the gNodeBAlgo.SSlotSchOpt parameter cannot be set to UL_SCHEDULE_OPT or DL_UL_SCHEDULE_OPT .

4.1.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Reverse spectrum scan	STR NRDUCELLRIC	None	After time-domain interference avoidance is enabled, only the GP symbols for time-domain interference avoidance can be scanned. Other GP symbols cannot be scanned, and consequently, the reverse spectrum scan data is invalid for these GP symbols.
4_1_SS4 adaptation	4_1_SS4_ADAPT_SW option of the gNodeBAlgo.UISchSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCell.SlotStructure parameter is set to SS4 , the following rule applies: If both time-domain interference avoidance and 4_1_SS4 adaptation are enabled, time-domain interference avoidance preferentially takes effect.
8_2_SS54 adaptation	8_2_SS54_ADAPT_SW option of the gNodeBAlgo.UISchSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCell.SlotStructure parameter is set to SS54 , the following rule applies: If both time-domain interference avoidance and 8_2_SS54 adaptation are enabled, time-domain interference avoidance preferentially takes effect.

4.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Cells

- The cell bandwidth is 20 MHz to 100 MHz, and the SCS is 30 kHz.
- The **NRDUCell.SlotAssignment** parameter is set to **8_2_DDDDDDDSUU**, **8_2_DDDSUUDDDD**, or **4_1_DDDSU**.
- The **NRDUCell.SlotStructure** parameter is set to **SS56**, **SS518**, **SS86**, **SS818**, **SS4**, **SS6**, **SS01001202**, **SS18**, or **SS54**.
- The **NRDUCell.FrequencyBand** parameter is not set to **N257**, **N258**, **N260**, or **N261**.
- The **NRDUCell.LampSiteCellFlag** parameter is set to **NO**, meaning that the cell is not a LampSite cell.

4.1.3.5 Others

None

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

[Table 4-2](#) describes the parameters used for function activation.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Time Interference Avoidance Switch	gNodeBAIgo.TimelntfrAvoidSw	• Select the SS56_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS56 and SS54, respectively. • Select the SS518_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS518 and SS54, respectively. • Select the SS86_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS86 and SS84, respectively. • Select the SS818_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS818 and SS84, respectively. • Select the SS4_2GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS4 and SS2, respectively. • Select the SS6_2GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS6 and SS2, respectively. • Select the SS6_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS6 and SS4, respectively. • Select the SS01001202_2GP_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS01001202 and SS2, respectively. • Select the SS01001202_4GP_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS01001202 and SS4, respectively.

Parameter Name	Parameter ID	Setting Notes
		• Select the SS18_2GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS18 and SS2, respectively. • Select the SS18_4GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS18 and SS4, respectively. • Select the SS54_2GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS54 and SS52, respectively. • Select the SS54_1GP_TIME_INTRF_AVOID_SW option if the self-contained slot structures of victim and interfering cells are SS54 and SS51, respectively.

4.1.4.1.2 Using MML Commands

NOTICE

- After time-domain interference avoidance is enabled, it takes effect in cells whose slot structures meet requirements. You are advised to enable this function when there is a small number of UEs. This is because this function will cause cells to automatically restart, causing service drops and re-access of admitted UEs.
- In LNR co-RF networking, NR cell establishment will fail if any of the following options is selected for the base station:
 - **SS54_2GP_TIME_INTRF_AVOID_SW**
 - **SS54_1GP_TIME_INTRF_AVOID_SW**
 - **SS86_4GP_TIME_INTRF_AVOID_SW**
 - **SS818_4GP_TIME_INTRF_AVOID_SW**
 - **SS56_4GP_TIME_INTRF_AVOID_SW**
 - **SS518_4GP_TIME_INTRF_AVOID_SW**

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Assume that the slot structure of an interfering base station is SS2 and that of a victim base station is SS4. In actual configuration, perform operations based on interference on live networks.  
MOD GNODEBALGO: TimeIntrfAvoidSw=SS4_2GP_TIME_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for time-domain interference avoidance  
MOD GNODEBALGO: TimeIntrfAvoidSw=SS4_2GP_TIME_INTRF_AVOID_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

- Run the **LST GNODEBALGO** command. If the **Time Interference Avoidance Switch** parameter has a value of **On** for any of its options in the command output, time-domain interference avoidance has been enabled.
- Observe the increase in the average uplink UE throughput (**User Uplink Average Throughput (DU)**) to check whether this function has taken effect.

4.1.4.3 Network Monitoring

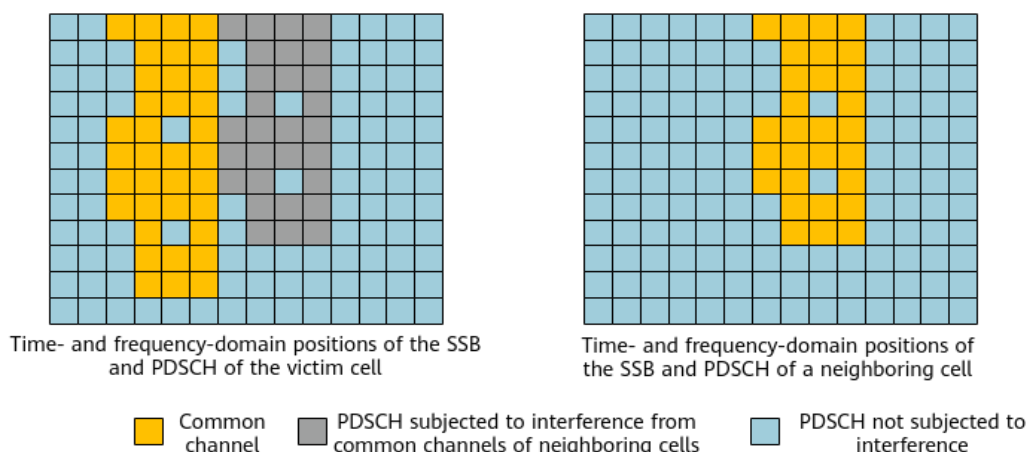
After this function takes effect, check the **User Uplink Average Throughput (DU)** for network monitoring.

4.2 Common Channel Interference Avoidance

4.2.1 Principles

When neighboring cells provide different quantities of synchronization signal and PBCH block (SSB) beams, the time- and frequency-domain positions where common channel data (including the SSB and RMSI) is transmitted cannot be completely aligned between the cells. (RMSI is short for Remaining Minimum SI.) In this case, common channels of neighboring cells cause interference to the physical downlink shared channel (PDSCH) of the local cell, severely affecting the downlink cell throughput, as shown in [Figure 4-3](#).

Figure 4-3 Common channel interference



To mitigate the impact of common channel interference on the NR network, common channel interference avoidance is introduced. After this function is enabled, **the base station checks whether a cell experiences interference from the common channels of neighboring cells. If such interference exists, downlink scheduling in the cell is not performed at the time-frequency positions available for the common channels of neighboring cells.** This function is controlled by the **SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter. It is recommended that this function be enabled when cells are configured with different quantities of SSB beams.

When the **NRDUCell.SsbTimePos** parameter is set to **SSB01_PCI_MOD2_STAGGER**, **SSB23_PCI_MOD2_STAGGER**, or **SSB_PCI_MOD3_STAGGER**, it is recommended that the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter be set to **INTRA_FREQ_SSB_INTRF_AVOID_SW** to reduce the interference from common channels of neighboring cells to the data channel of the serving cell.

For a fast track through the principles of common channel interference avoidance, see [\(Video\) Common Channel Interference Avoidance](#).



4.2.2 Network Analysis

4.2.2.1 Benefits

This function is used to stop the loss when inter-cell common channel interference occurs. In drive test (DT) scenarios, you are advised to enable this function if cells

are configured with different quantities of SSB beams and the downlink bit errors increase significantly in the slot where the SSBs of the neighboring cells are located.

The analysis of benefits in NR TDD cells is as follows:

- When full-buffer services are being performed:
 - The downlink UE throughput increases. For stationary UEs that are located at the cell center or at a medium distance from the cell center, the average downlink throughput increases by more than 10% if the PDSCH experiences interference from SSB transmitted in a neighboring cell.
 - For UEs in DTs, the increase in the average downlink throughput is related to the proportion of geographic locations where the UEs suffer SSB interference from neighboring cells in the DT routes. The larger the proportion, the higher the increase in the downlink throughput of UEs in DTs.
- When burst services are being performed, the gain of this function is related to the slot used for scheduling and cannot be specifically determined, but there is no negative impact.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

This function reduces the available time-frequency resources for downlink scheduling while avoiding common channel interference. The reduction (expressed in percentage) of time-frequency resources available for downlink scheduling is related to the cell bandwidth and the number of SSB resources in neighboring cells. A smaller cell bandwidth and a larger number of SSB resources in neighboring cells result in a larger reduction of time-frequency resources available for downlink scheduling. For example, with typical bandwidths, if full buffer services are performed in a neighboring cell that has the maximum number of SSB resources, the reduction (expressed in percentage) of time-frequency resources available for downlink scheduling is as follows:

- When the bandwidth is 40 MHz, the available RBs decrease by up to 8%, in which case the **Cell Downlink Average Throughput (DU)** decreases by up to 8%.
- When the bandwidth is 80 MHz, the available RBs decrease by up to 4%, in which case the **Cell Downlink Average Throughput (DU)** decreases by up to 4%.
- When the bandwidth is 100 MHz, the available RBs decrease by up to 3%, in which case the **Cell Downlink Average Throughput (DU)** decreases by up to 3%.

4.2.3 Requirements

4.2.3.1 Licenses

None

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Blanking	NRDUCell.PeerCellSlotAssignment	<i>Standard Compliance</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellSlotAssignment parameter to a value other than NOT_CONFIG), common channel interference avoidance cannot be enabled.

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station DL CoMP is enabled, common channel interference avoidance does not take effect for CoMP UEs, but it can still take effect for other types of UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Load-based adaptive downlink scheduling	NRDUCellDLSch.DLSchPolicy set to ADAPTIVE	<i>Downlink Scheduling</i>	Common channel interference avoidance decreases the number of available downlink RBs in a cell. As a result, the downlink PRB usage may be lower than Downlink Schedule Heavy Load PRB Threshold (which is used for determining whether load-based adaptive downlink scheduling can work). If this occurs, load-based adaptive downlink scheduling can provide benefits in fewer scenarios.
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnlIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	Only enhanced intra-frequency SSB interference avoidance takes effect if this function and common channel interference avoidance are both enabled.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Assume that cell B is a neighboring cell of cell A. If the RMSI broadcast interval of cell A is adjusted, interference may occur between the PDSCH of cell A and common messages of cell B, decreasing the downlink throughput of cell A. It is therefore recommended that common channel interference avoidance be enabled in both cell A and cell B.

4.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Others

The Xn interface between gNodeBs must be normal. (If no Xn interface is configured, this function will trigger automatic Xn interface setup.)

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChannelIntrfAvoidSwitch	SSB_INTRF_AVOID_SW	It is recommended that this option be selected when adjacent cells are configured with different quantities of SSBs.

Parameter Name	Parameter ID	Option	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch	INTRA_FREQ_SSB_INTRF_AVOID_SW	It is recommended that this option be selected when NRDUCell.SsbTimePos is set to SSB01_PCI_MOD2_STAGGER , SSB23_PCI_MOD2_STAGGER , or SSB_PCI_MOD3_STAGGER .

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling common channel interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=SSB_INTRF_AVOID_SW-1;
//Enabling intra-frequency SSB interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CommChnIntrfAvoidSwitch=INTRA_FREQ_SSB_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling common channel interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=SSB_INTRF_AVOID_SW-0;
//Disabling intra-frequency SSB interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CommChnIntrfAvoidSwitch=INTRA_FREQ_SSB_INTRF_AVOID_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

When a neighboring cell and serving cell have a different number of configured SSB beams, this function takes effect if no scheduling is performed in the serving cell at the time-frequency positions where SSBs are only sent in the neighboring cell.

 NOTE

The number of SSB beams is indicated by the **ssb-PositionInBurst** field in the **ServingCellConfigCommonSIB** message. For details, see 3GPP TS 38.331.

4.2.4.3 Network Monitoring

After this function is enabled, observe the change in the following KPI to determine whether it has taken effect:

User Downlink Average Throughput (DU), which is expected to increase

5

Intra-Frequency Interference in LTE and NR Co-Coverage Scenarios

In LTE and NR co-coverage scenarios with the spectrum of an LTE cell overlapping that of an NR cell, the LTE cell causes intra-frequency interference to the NR cell. Such interference is referred to as intra-frequency interference in LTE and NR co-coverage scenarios.

5.1 Interference-based Subband Robust Weights (Low-Frequency TDD)

5.1.1 Principles

According to 3GPP specifications, LTE FDD Band 7 and NR TDD n41 are available in 2.6 GHz, as described in [Table 5-1](#). When LTE FDD Band 7 and NR TDD n41 coexist, the downlink of the former will cause interference to the uplink of the latter. In the view of time-domain interference (shown in [Figure 5-1](#)), both self-contained slots and uplink slots in NR suffer interference from the downlink of LTE FDD. In the view of frequency-domain interference (shown in [Figure 5-2](#)), interference exists in the frequency range from 2620 MHz to 2690 MHz.

Table 5-1 Frequency bands in 2.6 GHz

Band	Uplink Frequency Range	Downlink Frequency Range	Duplex Mode
Band 7	2500 MHz to 2570 MHz	2620 MHz to 2690 MHz	FDD
n41	2496 MHz to 2690 MHz	2496 MHz to 2690 MHz	TDD

Figure 5-1 Time-domain interference characteristics

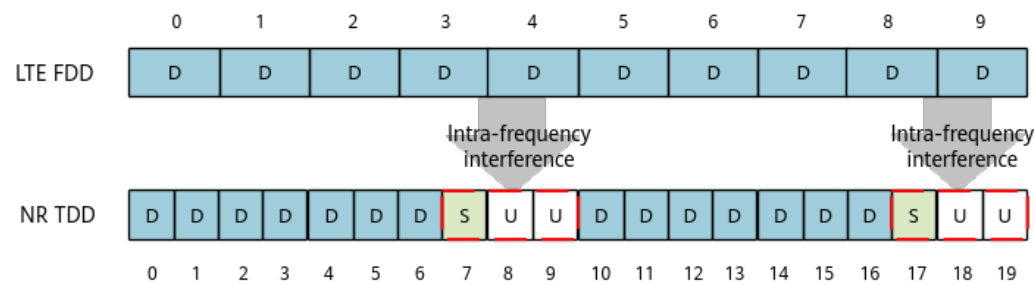
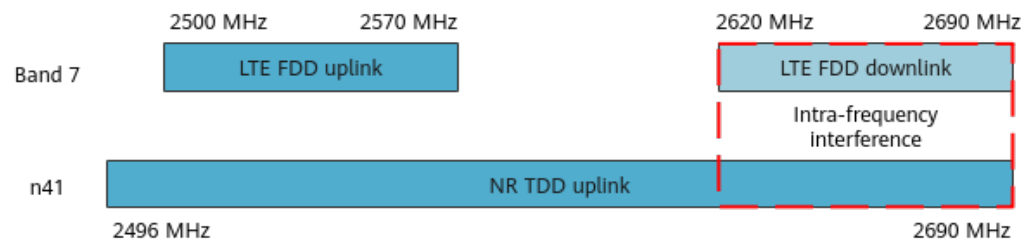


Figure 5-2 Frequency-domain interference characteristics



As mentioned above, both self-contained slots and uplink slots suffer interference in NR TDD. Therefore, interference is inevitably caused to SRSs. In case of strong interference to SRSs, UEs that rely on downlink SRS-based weights will experience decreased performance. When there is interference to SRSs, the interference-based subband robust weights function can be used to improve the performance of UEs that rely on SRS-based weights. With this function, **a subband robust weight is calculated based on SRS measurements over a non-interference range and is used across the entire band**. Weight robustness is improved owing to channel filtering in SRS frequency hopping subbands. It should be noted that a subband robust weight applies to a scenario where SRS measurements do not cover the entire band.

This function is controlled by the **INTRF_BASED_ROBUST_WEIGHT_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter. The detailed process of this function is as follows:

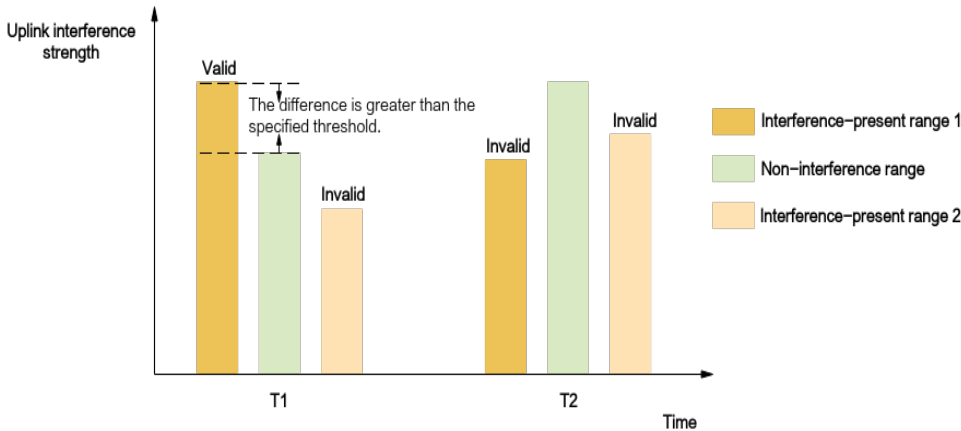
- 1. Interference range determination
 - a. Manual configuration of an uplink interference range

Configura tion	Parameter	Description
Adding an interferenc e range index	NRDUCellIntrfBand.Int erferenceBandIndex	Up to two uplink interference ranges can be configured.
Configurin g an interferenc e range	NRDUCellIntrfBand.Sta rtRb and NRDUCellIntrfBand.En dRb	The parameters are separately used to configure the start RB position and end RB position for each interference range.

Configura tion	Parameter	Description
Configurin g the interferenc e direction for an interferenc e range	NRDUCellIntrfBand.Int erferenceDirection	This parameter must be set to UPLINK (indicating uplink interference) or UPLINK_DOWNLINK (indicating both uplink and downlink interference). Otherwise, this function will not take effect.

- b. Interference range validity determination
- The base station compares the difference in uplink interference strength between the interference and non-interference ranges with the sum of the **NRDUCellIntrfIdent.UlIntrfDiffThld** and **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter values every 5 minutes. If the difference is greater than the sum, the interference range is considered to be valid; otherwise, the interference range is considered to be invalid. Weight switching can be performed only when there is a valid interference range. [Figure 5-3](#) illustrates interference range validity determination.

Figure 5-3 Interference range validity determination



2. Weight switching
- The weight for a UE is switched to a robust weight when either of the following conditions is met:
- The SRS period is greater than or equal to **NRDUCellPdschPrecode.RobustWtFullbandSrsPrdThld** and the channel correlation is less than **NRDUCellPdschPrecode.IntrfBasedRobustWtRhoThld**.
 - The SRS preSINR is less than or equal to **NRDUCellPdschPrecode.RobustWtSrsPresinrThld** and the channel correlation is less than **NRDUCellPdschPrecode.IntrfBasedRobustWtRhoThld**.

This function takes effect only for 4-port UEs supporting SRS antenna selection. It does not take effect if any of the following conditions is met:

- The full band is configured as an interference range, or a complete SRS frequency hopping over the non-interference frequency range cannot be achieved due to a too large interference range.
- Channel calibration fails due to strong interference.

If a too large interference range causes this function not to take effect, you can reconfigure the **NRDUCellSrs.SrsWideBandIndexCsrs** parameter to increase the SRS frequency hopping multiplier so that at least a complete SRS frequency hopping over the non-interference frequency range can be achieved and this function can take effect. For details about SRS, see descriptions of SRS resource management in *Channel Management*.

5.1.2 Network Analysis

5.1.2.1 Benefits

Deployment suggestion: Enable this function in a scenario where a certain frequency range of an NR TDD cell suffers interference from LTE FDD Band 7 and the value of **N.UL.NI.Avg** (used to measure the average interference and noise received on each uplink PRB) is greater than -105 dBm.

Gains: After this function is enabled in the aforementioned scenario, the **User Downlink Average Throughput (DU)**, which indicates the average downlink UE throughput, increases. When the value of **N.UL.NI.Avg** is over 20 dB greater than the noise floor (-117 dBm/RB), the average downlink UE throughput increases by 10–20%. Note that the gains obtained when MetaAAU modules are used are relatively low as MetaAAUs themselves have a strong anti-interference capability.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

A prerequisite for this function to take effect is that at least a complete SRS frequency hopping over the non-interference frequency range can be achieved. If the number of SRS frequency hops increases due to the SRS frequency hopping configuration being modified, the following impacts are caused:

- The period for full-bandwidth SRS measurements is prolonged. Consequently, UE performance deteriorates in mobility scenarios but the SRS signal quality improves.
- The total number of measurements increases. Consequently, the number of frequency ranges with poor SINR increases if strong interference exists on a partial bandwidth of a cell. As a result, the counters related to results of contiguous SRS measurements may be affected. An example of such counters is **N.NsaDc.ULChangetoMeNB** (used to measure the number of times the uplink data transmission of NSA DC UEs is transferred to the MeNB in a cell).

5.1.3 Requirements

5.1.3.1 Licenses

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-081208	Inter-Mode Interference Suppression	NR0S00RA RA00	gNodeB	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
UE-characteristic-based SRS period adaptation	USER_CHARACTERISTIC_SRS_ADAPT_SWITCH option of the NRDUCellSrs.SrsAdaptSwitch parameter	<i>Channel Management</i>	The interference-based subband robust weights function can be enabled only when UE-characteristic-based SRS period adaptation is enabled.

Function Name	Function Switch	Reference	Description
DL MU NLOS IR Precoding Low- Complexity Sw	DL_MU_NLOS_IR_PREC_LOW_CPLX_SW option of the NRDUCellAlgoSwitch . <i>MuMimoSwitch</i> parameter	None	In scenarios where the DL_MU_NLOS_IR_PRECODING_SW option of the NRDUCellAlgoSwitch . <i>MuMimoSwitch</i> parameter is selected, interference-based subband robust weights can be enabled only when both of the following conditions are met: • The DL_MU_NLOS_IR_PREC_LOW_CPLX_SW option of the NRDUCellAlgoSwitch.<i>MuMimoSwitch</i> parameter is selected. • The NRDUCellPdschPrecode.<i>DLIMuIRPrecodePol</i> parameter is set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER_POWER_ENH.
DL MU Intra- Reject Precoding Policy	NRDUCellPdschPrecode . <i>DLIMuIRPrecodePol</i> set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER_POWER_ENH	None	

5.1.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
SRS Narrowband Index Csrs	NRDUCellSrs . <i>SrsNarrowBandIndexCsrs</i>	None	When the interference-based subband robust weights function is enabled, the NRDUCellSRS . <i>SrsNarrowBandIndexCsrs</i> parameter cannot be set to 52 , 53 , 57 , or 61 .

Function Name	Function Switch	Reference	Description
SRS Wideband Index Csrs	NRDUCellSrs.SrsWideBandIndexCsrs	None	When the interference-based subband robust weights function is enabled, the NRDUCellSrs.SrsWideBandIndexCsrs parameter cannot be set to 52, 53, 57, or 61 .
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Cell Combination is mutually exclusive with interference-based subband robust weights.
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper Cell is mutually exclusive with interference-based subband robust weights.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Fusion Cell is mutually exclusive with interference-based subband robust weights.

5.1.3.2.3 Function Impacts

Function	Function Switch	Reference	Description
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	The interference-based subband robust weights function does not take effect for UEs that report event A3 for CoMP.
Low-complexity measurement of SRS-based weights	SRS_WT_MEAS_LOW_CPLX_SW option of the NRDUCellDLAmc.DLMcsSelAlgoSw parameter	None	When enabled together with 'low-complexity measurement of SRS-based weights', 'interference-based subband robust weights' takes effect preferentially.

Function	Function Switch	Reference	Description
External interference avoidance	UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent. IntraFrequencyInterferenceAvoidanceMethod parameter NRDUCellIntrfIdent. IntraFrequencyInterferenceAvoidanceMethod	<i>Interference Avoidance</i>	When enabled together with external interference avoidance, the 'interference-based subband robust weights' function does not take effect.
Load-based adaptive downlink scheduling	NRDUCellDLISch.DLISchPolicy set to ADAPTIVE	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When both 'interference-based subband robust weights' and 'load-based adaptive downlink scheduling' are enabled, only 'load-based adaptive downlink scheduling' takes effect.
Adaptive filtering for SRS channel estimation	CHN_EST_ADAPT_FILTER_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Adaptive filtering for SRS channel estimation will yield fewer benefits when it is enabled together with interference-based subband robust weights.
Self-fusion weights	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with interference-based subband robust weights.

5.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

[5.1.4.1.1 Data Preparation](#) describes the parameters used for function activation.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	To enable interference-based subband robust weights, select the INTRF_BASED_ROBUST_WEIGHT_SW option.
Interference Band Index	NRDUCellIntrfBand.InterferenceBandIndex	Set this parameter based on the frequency-domain position where interference occurs. A maximum of two interference ranges can be configured.
Start RB	NRDUCellIntrfBand.StartRb	It is recommended that this parameter be set based on the actual interference range.
End RB	NRDUCellIntrfBand.EndRb	It is recommended that this parameter be set based on the actual interference range.
Interference Direction	NRDUCellIntrfBand.InterferenceDirection	Set this parameter to UPLINK or UPLINK_DOWNLINK .

Parameter Name	Parameter ID	Setting Notes
UL Interference Difference Threshold	NRDUCellIntrfIdent.UlIntrfDiffThld	It is recommended that the sum of the NRDUCellIntrfIdent.UlIntrfDiffThld and NRDUCellIntrfIdent.DlIntrfIdentOfs parameter values be 5 dB.
DL Interference Identification Offset	NRDUCellIntrfIdent.DlIntrfIdentOfs	
Robust Weight SRS PreSINR Thld	NRDUCellPdschPre-code.RobustWtSrsPresinrThld	Use the recommended value.
Robust Weight Fullband SRS Period Thld	NRDUCellPdschPre-code.RobustWtFullbandSrsPrdThld	Use the recommended value.
DL MCS Selection Algorithm Switch	NRDUCellDLAMC.DlMcsSelAlgoSw	Select the RANK_ADAPT_MCS_ADJ_SW option.
DL Rank Adaptation MCS Threshold	NRDUCellDLRANK.DlRankAdaptMcsThld	Use the recommended value.
Robust Weight Rank1 SE Factor	NRDUCellDLRANK.RobustWeightRank1Factor	Use the recommended value.
Robust Weight Phase Calculation Method	NRDUCellPDSCH.RobustWtPhaseCalcMethod	Set this parameter to SRS_H_BASED_CALCULATION .
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDLMIMO.DLMuMimoSchSupplementSw	It is recommended that the ROBUST_WT_UE_LOOSE_PAIR_SW option be deselected.
Intrf Based Rho Value Thld for Robust Weight	NRDUCellPdschPre-code.IntrfBasedRobustWtRhoThld	Use the recommended value. In strong interference scenarios, the threshold specified by this parameter can be increased so that the involved function can take effect more easily.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling interference-based subband robust weights
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=INTRF_BASED_ROBUST_WEIGHT_SW-1;
//Manually configuring an uplink interference range (70 MHz, which is the maximum interference range in
LTE FDD Band 7, is used as an example)
ADD NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0, StartRb=55, EndRb=244,
InterferenceDirection=UPLINK;
//Configuring UL Interference Difference Threshold and DL Interference Identification Offset
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, UlIntrfDiffThld=5, DlIntrfIdentOfs=0;
//Configuring thresholds related to robust weights
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=101,RobustWtSrsPresInrThld=-8,
RobustWtFullbandSrsPrdThld=960, IntrfBasedRobustWtRhoThld=70;
//Selecting the RANK_ADAPT_MCS_ADJ_SW option
MOD NRDUCELLDLAMC: NrDuCellId=101, DlMcsSelAlgoSw=RANK_ADAPT_MCS_ADJ_SW-1;
//Setting RobustWeightRank1Factor and DlRankAdaptMcsThld
MOD NRDUCELLDLRANK: NrDuCellId=101, DlRankAdaptMcsThld=9, RobustWeightRank1Factor=80;
//Setting RobustWtPhaseCalcMethod
MOD NRDUCELLPDSCH: NrDuCellId=101, RobustWtPhaseCalcMethod=SRS_H_BASED_CALCULATION;
//Deselecting the ROBUST_WT_UE_LOOSE_PAIR_SW option
MOD NRDUCELLDLMIMO:
NrDuCellId=101,DLMuMimoSchSupplementSw=ROBUST_WT_UE_LOOSE_PAIR_SW-0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing an uplink interference range
RMV NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0;
//Disabling interference-based subband robust weights
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=INTRF_BASED_ROBUST_WEIGHT_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

If the **User Downlink Average Throughput (DU)** increases after this function is enabled, this function has taken effect.

5.1.4.3 Network Monitoring

The **User Downlink Average Throughput (DU)** is expected to increase.

6 Common Interference Avoidance Technologies

Common interference avoidance technologies are used for interference avoidance irrespective of interference types.

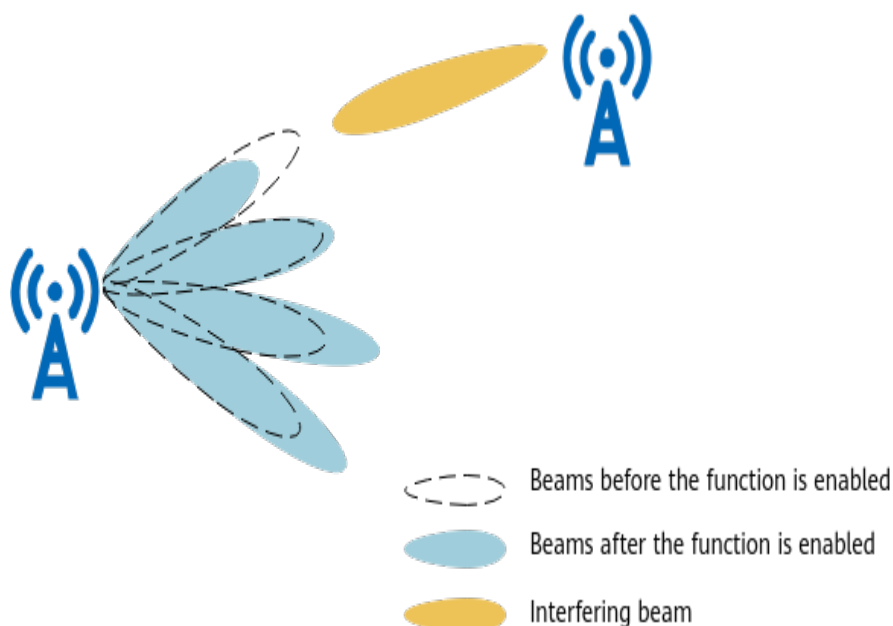
6.1 Frequency Offset Measurement Optimization Based on Interference Suppression (Low-Frequency TDD)

6.1.1 Principles

To improve network performance when there is interference from neighboring cells or other systems, Huawei has introduced frequency offset measurement optimization based on interference suppression. After this function is enabled, **the gNodeB performs interference suppression by weakening the coverage of receive beams in the direction of interference but strengthening the coverage in other directions, during the estimation of the uplink frequency offset. This improves estimation accuracy**, and increases the uplink and downlink system capacity as well as the user-perceived rate under light to moderate network load. [Figure 6-1](#) shows how this function works.

This function is controlled by the **INTRF_FREQOFFSET_OPT_SW** option of the **NRDUCellPusch.PuschMeasSw** parameter.

Figure 6-1 Frequency offset measurement optimization based on interference suppression



6.1.2 Network Analysis

6.1.2.1 Benefits

- In scenarios with interference and light network load, frequency offset measurement optimization based on interference suppression increases the uplink user-perceived rate and average uplink cell throughput.
- In scenarios with heavy load and scenarios without interference, this function does not generate negative impacts but may deliver no gains.

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

There are no impacts on the network.

6.1.3 Requirements

6.1.3.1 Licenses

None

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

None

6.1.3.2.2 Mutually Exclusive Functions

None

6.1.3.2.3 Function Impacts

Function	Function Switch	Reference	Description
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	After 'frequency offset measurement optimization based on interference suppression' takes effect, the average downlink cell throughput may increase in scenarios where DL CoMP is enabled.
Co-MM	INTRA_GNB_SU_COM_M_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	After 'frequency offset measurement optimization based on interference suppression' takes effect, the average downlink cell throughput may increase in scenarios where Co-MM is enabled.
Interference-based precise frequency offset estimation	INTRF_PREC_FREQ_OFFSET_EST_SW option of the NRDUCellPusch.PuschMeasSw parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Frequency offset measurement optimization based on interference suppression does not take effect when it is enabled together with interference-based precise frequency offset estimation.

6.1.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.1.3.4 Cells

The **NRDUCellTrp.TxRxMode** parameter cannot be set to **1T1R**.

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Measurement Sw	NRDUCellPusch.PuschMeasSw	INTRF_FREQOFFSET_OPT_SW	Select this option.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to **6.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling frequency offset measurement optimization based on interference suppression  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=INTRF_FREQOFFSET_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling frequency offset measurement optimization based on interference suppression  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=INTRF_FREQOFFSET_OPT_SW-0;
```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

- Run the **LST NRDUCELLPUSCH** command. If the **PUSCH Measurement Sw** parameter has a value of **Interference-based Freq Ofs Meas Opt Sw:On** in the command output, frequency offset measurement optimization based on interference suppression has been enabled.
- If the **Cell Uplink Average Throughput (DU)** increases after frequency offset measurement optimization based on interference suppression is enabled, this function has taken effect.

6.1.4.3 Network Monitoring

Observe the change in the following indicator:

Cell Uplink Average Throughput (DU), which is expected to increase

6.2 External Interference Avoidance (Low-Frequency TDD)

6.2.1 Principles

Live networks are subjected to different external interference, with the majority of such interference being uplink interference from non-NR systems. Due to varying interference sources, the interference exhibits different time-frequency characteristics.

- The interference that originates from a fixed source (for example, the interference from a persistent source) exhibits the following time-frequency characteristics:
 - Stable in the frequency domain
 - Continuous in the time domain
 - Relatively fixed bandwidth over which interference is present
- The interference that originates from non-fixed sources (for example, the interference from multiple sources in an area or the interference from a source does not always exist) exhibits the following time-frequency characteristics:
 - Non-stable in the frequency domain, with the victim frequency domain being fixed in a cell but different between cells within an area
 - Discontinuous in the time domain, with the interference strength changing significantly with time
 - Non-fixed bandwidth over which interference is present

Against this background, the external interference avoidance function is introduced to mitigate the impacts of external interference on an NR network. **This function achieves interference avoidance during scheduling. The decision on whether to enable this function is based on interference. O&M personnel can evaluate interference on the live network and determine frequency-domain positions subjected to interference based on the average interference and noise detected on the uplink full-bandwidth PRBs of a cell (for example, indicated by counters N.UL.NI.Avg.PRBO to N.UL.NI.Avg.PR272 for a cell with a 100 MHz bandwidth) or by using the cell interference tracing function of the MAE or other interference evaluation tools.**

This function is controlled by the **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter and the **UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter.

Figure 6-2 Process of external interference avoidance

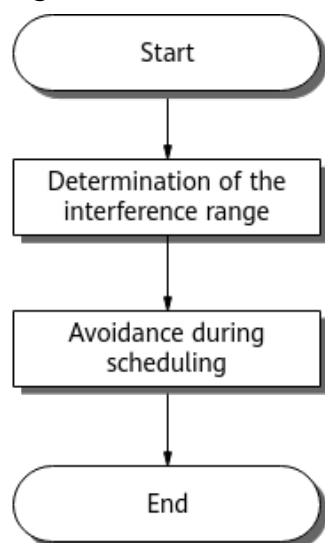


Figure 6-2 shows the process of external interference avoidance. Details are as follows:

1. Interference range determination

If interference originates from fixed sources, manual configuration of an uplink interference range is recommended. Otherwise, automatic identification of an uplink interference range is used.

– Manual configuration of an uplink interference range

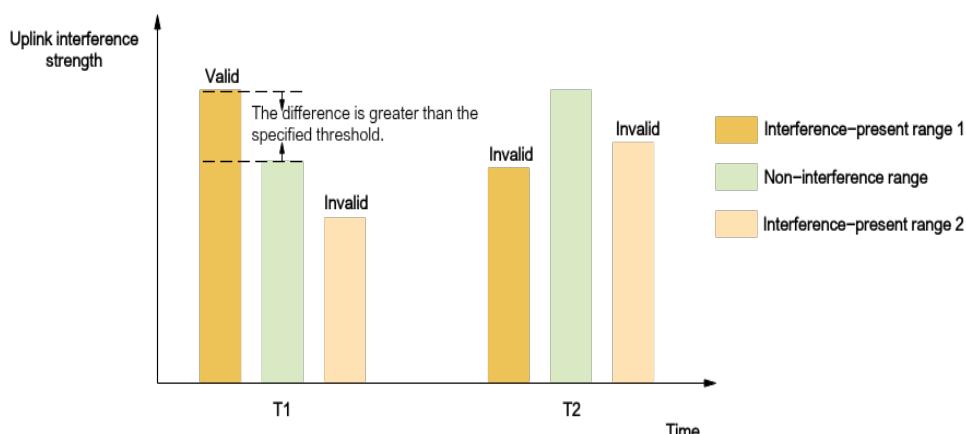
i. Manual configuration of an uplink interference range

Configuration	Parameter	Description
Selecting manual configuration of an uplink interference range	NRDUCellIntrfIdent.IntrfIdentificationMethod	Set this parameter to UL_MANUAL .
Adding an interference range index	NRDUCellIntrfBand.InterferenceBandIndex	Up to two interference ranges can be configured.
Configuring an interference range	NRDUCellIntrfBand.StartRb and NRDUCellIntrfBand.EndRb	The parameters are separately used to configure the start RB position and end RB position for each interference range.
Configuring the interference direction for an interference range	NRDUCellIntrfBand.InterferenceDirection	Set this parameter to UPLINK , indicating uplink interference.

ii. Interference range validity determination

After an interference range is manually configured, the base station compares the difference in uplink interference strength between the interference and non-interference ranges with the **NRDUCellIntrfIdent.UlIntrfDiffThld** parameter value every 5 minutes. If the difference is greater than this parameter value, the interference range is considered to be valid; otherwise, the interference range is considered to be invalid. **Figure 6-3** illustrates interference range validity determination. For a valid interference range, avoidance is needed during scheduling. Conversely, for an invalid interference range, avoidance is unneeded during scheduling.

Figure 6-3 Interference range validity determination



- Automatic identification of an uplink interference range

The **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter can be set to **UL_AUTO** to enable automatic identification of an uplink interference range. The base station performs interference range identification every 5 minutes. The base station identifies a frequency domain range as an uplink interference range when the measurement results of the frequency domain range meet the following requirement:

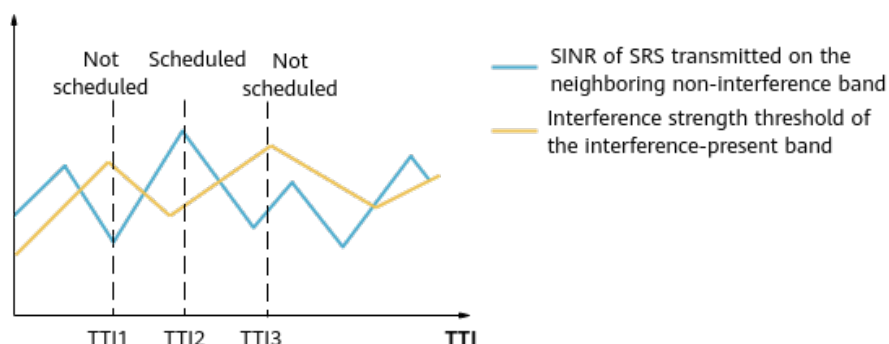
Uplink interference strength measured in the frequency domain range > Average full-band uplink interference strength + Offset to the threshold for automatic detection of uplink interference
(**NRDUCellIntrfIdent.UlIntrfAutoDetectThldOfs**)

2. Avoidance during scheduling

The base station performs the following scheduling processing in the identified interference range:

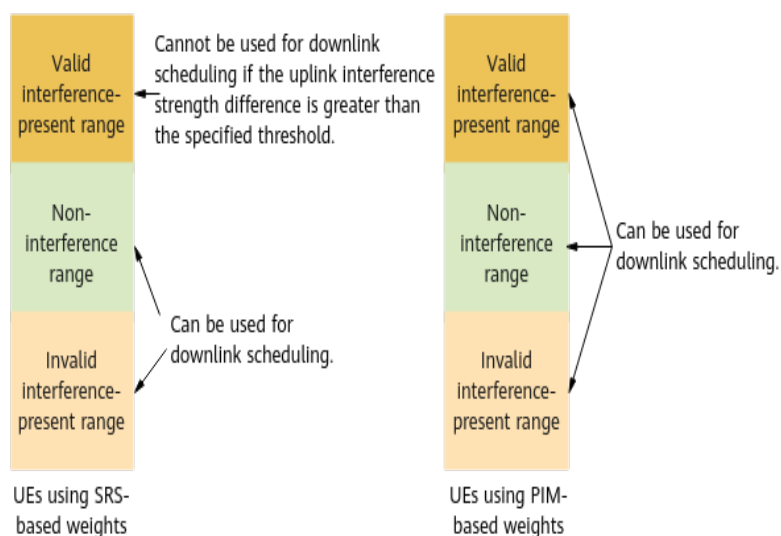
- Uplink scheduling (**Figure 6-4**): Whether the interference range can be used for uplink scheduling depends on the relationship between the SINR of SRS transmitted on the neighboring non-interference range and the interference strength threshold of the interference range (this threshold is related to **NRDUCellIntrfIdent.UlIntrfFreqSchJudgeOfs**). Details are as follows:
 - If the SINR of SRS transmitted on the neighboring non-interference range is greater than this interference strength threshold, the interference range can be used for uplink scheduling.
 - In other cases, the interference range cannot be used for uplink scheduling.

Figure 6-4 Avoidance during uplink scheduling



- Downlink scheduling (**Figure 6-5**):
 - For UEs that rely on SRS-based weights:
 - If the uplink interference range is manually configured, the following mechanism applies: If the difference in uplink interference strength between the interference and non-interference ranges is greater than the sum of the **NRDUCellIntrfIdent.UlIntrfDiffThld** and **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter values, downlink scheduling is not performed in the interference range; otherwise, downlink scheduling can be performed in the interference range.
 - If the uplink interference range is automatically identified, the following mechanism applies: If the uplink interference strength measured in the interference range is greater than the average full-bandwidth uplink interference strength plus the sum of the **NRDUCellIntrfIdent.UlIntrfAutoDetectThldOfs** and **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter values, downlink scheduling is not performed in the interference range; otherwise, downlink scheduling can be performed in the interference range.
 - For UEs that rely on PMI-based weights, downlink SU-MIMO scheduling and beam-domain pairing for downlink MU-MIMO are allowed in the interference range, but correlation-based UE pairing for downlink MU-MIMO is not supported.

Figure 6-5 Avoidance during downlink scheduling



For details about SRS- and PMI-based weights, see "Downlink Beamforming" in *MIMO (TDD)*.

If cell performance deteriorates due to interference on the PUCCH and uplink interference ranges are manually configured, the **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter can be selected to enable PUCCH interference avoidance. After this option is selected, **the PUCCH is not transmitted in the frequency range where interference is present**. As a result, the PUSCH frequency-domain positions may be discontinuous. If there are UEs that do not support non-consecutive PUSCH scheduling on the network, the uplink peak rates of these UEs will decrease.

If UE access is affected as a result of deterioration of SSB or BWP0 performance due to uplink external interference, the SSB center frequency can be manually configured to avoid interference.

- When CORESET 0 is configured to contain 24 RBs (that is, the **NRDUCellCoreset.CommonCtrlResRbNum** parameter is set to **RB24**), the configured SSB center frequency cannot be in the range of **[NRDUCellIntrfBand.StartRb – 18 RBs, NRDUCellIntrfBand.EndRb + 15 RBs]**. For details about the SSB center frequency, see "SSB Frequency-Domain Position" in *Cell Management*.
- When CORESET 0 is configured to contain 48 or 96 RBs (that is, the **NRDUCellCoreset.CommonCtrlResRbNum** parameter is set to **RB48** or **RB96**), the configured SSB center frequency is 27 RBs beyond the start and end RBs of the interference range in the n40/n41/n77/n78 band, and 39 RBs beyond the start and end RBs of the interference range in the n79 band.

Once the SSB center frequency is manually configured to avoid interference, neighbor relationships will be changed. (One example of this change is that intra-frequency neighboring cells will be changed to inter-frequency neighboring cells.) In this case, neighbor relationships and frequencies for inter-frequency handovers need to be reconfigured based on the new frequency position. If UEs in a cell do not support inter-frequency handovers, handovers will fail. If the SSB center frequency cannot be changed on live networks, automatic identification of an

uplink interference range can be used by setting the **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter to **UL_AUTO**.

For a fast track through the principles of external interference avoidance, see [\(Video\) External Interference Avoidance](#).



6.2.2 Network Analysis

6.2.2.1 Benefits

This function stops the loss when a cell experiences external interference. It is recommended that this function be enabled when the interference strength in the uplink interference range is over 20 dB greater than the average interference strength in the full band and the proportion of the uplink interference range to the available band for scheduling is less than 25%.

When full buffer services are running, enabling this function improves the average uplink throughput of stationary UEs at the cell edge or at a medium distance from the cell center by over 5%. However, no gains are generated if scheduling of UEs at the cell center continues in the interference range. In drive test scenarios, the average uplink throughput of UEs improves by 2–3%.

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

- Interference avoidance leads to a decrease in the available bandwidths allocated to uplink and downlink data channels. The proportion of decreased bandwidths is related to the configured interference bandwidth. Available RB resources decrease by 25% maximally.
- The values of **N.S.Symbol10.NI.Avg**, **N.S.Symbol11.NI.Avg**, **N.S.Symbol12.NI.Avg**, and **N.S.Symbol13.NI.Avg** fluctuate with scheduling of UEs and may present an overall decreasing trend.

6.2.3 Requirements

6.2.3.1 Licenses

None

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

None

6.2.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
PUCCH interference coordination	PUCCH_INTRF_COORD_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>MIMO (TDD)</i>	PUCCH interference coordination cannot be enabled when all the following conditions are met: • The PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter is selected. • The UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_DL_MANUAL. • The BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is deselected.

Function Name	Function Switch	Reference	Description
			PUCCH interference coordination cannot be enabled when all the following conditions are met: • The PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter is selected. • The UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_MANUAL. • The BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is deselected.

Function Name	Function Switch	Reference	Description
Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3	FORMAT1_FREQ_INTRF_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 cannot be enabled when all of the following conditions are met: • PUCCH interference avoidance (controlled by the PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter) is enabled. • Uplink cell-level interference avoidance (controlled by the UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. • Downlink UE-level interference avoidance (controlled by the DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled. • The NRDUCellIntrfIdent.IntrfIdentificationMethod

Function Name	Function Switch	Reference	Description
			parameter (specifying the interference identification method) is set to UL_DL_MANUAL. • The power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrS aving.BwpPwrSavi ngSw parameter) is disabled. Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 cannot be enabled when all of the following conditions are met: • PUCCH interference avoidance (controlled by the PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter) is enabled. • Uplink UE-level interference avoidance (controlled by the UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) is enabled.

Function Name	Function Switch	Reference	Description
			• The NRDUCellIntrflde nt.IntrflIdentifica- tionMethod parameter (specifying the interference identification method) is set to UL_MANUAL. • The power saving BWP (controlled by the BWP2_SWITCH option of the NRDUCellUePwrS aving.BwpPwrSavi ngSw parameter) is disabled.
Downlink multi-BWP interference avoidance	DL_BWP_HYBRID_IN TRF_RANDOM_SW option of the NRDUCellDISch.DlSc hAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink multi-BWP interference avoidance must be disabled when all the following conditions are met: • Interference ranges are manually configured for both uplink and downlink interference identification. • Cell-level interference avoidance is used in the uplink. • UE-level interference avoidance is used in the downlink.

Function Name	Function Switch	Reference	Description
Cell-level PUSCH interference avoidance	• UL_CELL_LEVEL_I NTRF_AVOID and DL_UE_LEVEL_INT RF_AVOID options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter both selected • NRDUCellIntrfIdent.IntrfIdentificationMethod parameter set to UL_DL_MANUAL • At least one uplink interference range for which the NRDUCellIntrfBand.InterferenceDirection parameter is set to UPLINK	<i>Interference Avoidance</i>	None

6.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If NRDUCellIntrfIdent./ntrfIdentificationMethod is set to UL_MANUAL , the following impact arises: BWP2 UEs may lack the necessary resources for scheduling if the BWP2 frequency range overlaps with the interference-affected range. If the UEs still lack the necessary resources for scheduling after several attempts by the base station to lower the rate threshold, they will switch to BWP1, affecting their power saving performance.
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.h.CompSwitch parameter	<i>CoMP</i>	After external interference avoidance is enabled, no SRS is transmitted in the interference-affected range. As a result, the proportion of occasions where DL CoMP takes effect decreases.
Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.h.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	After external interference avoidance is enabled, no SRS is transmitted in the interference-affected range. As a result, the proportion of occasions where coordinated interference management takes effect decreases.

Function Name	Function Switch	Reference	Description
Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	When the interference-affected range conflicts with the fixed resources reserved for services requiring connectivity with high reliability, you are advised not to reserve fixed resources for such services.
Co-MM	INTRA_GNB_SU_CO_MM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	After external interference avoidance is enabled, no SRS is transmitted in the interference-affected range. As a result, the probability that Co-MM takes effect decreases.
Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	When external interference avoidance and the interference-based subband robust weights function are both enabled, the latter function does not take effect.

Function Name	Function Switch	Reference	Description
Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailBandwidth > 20 MHz	<i>RedCap</i>	For an NR TDD cell, the number of activated RedCap BWP1s may be less than the number of BWP1s corresponding to the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter value when all of the following conditions are met: • PUCCH interference avoidance (controlled by the PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter) is enabled. • External interference avoidance is enabled. • At least one NRDUCellIntrfBand MO instance has been configured for the cell (that is, the command output of LST NRDUCELLINTRFBAND shows that there is at least one interference band).

6.2.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules that work in n40, n41, n77, n78, or n79 support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.2.3.4 Cells

The cell bandwidth must be 40 MHz/50 MHz/60 MHz/80 MHz/90 MHz/100 MHz.

6.2.3.5 Others

The number of RBs that can be configured for each interference range varies with the cell bandwidth. Details are as follows:

- If the bandwidth is 40 MHz, each interference range can contain up to 14 RBs.
- If the bandwidth is 50 MHz, each interference range can contain up to 27 RBs.
- If the bandwidth is 60 MHz, each interference range can contain up to 40 RBs.
- If the bandwidth is 80 MHz, each interference range can contain up to 54 RBs.
- If the bandwidth is 90 MHz, each interference range can contain up to 54 RBs.
- If the bandwidth is 100 MHz, each interference range can contain up to 68 RBs.

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

[Table 6-2](#) and [Table 6-3](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Interference Optimization Method	NRDUCellIntrfIdent.IntrfOptimizationMethod	Select the UL_UE_LEVEL_INTRF_AVOID option and deselect the other options.

Parameter Name	Parameter ID	Setting Notes
Interference Identification Method	NRDUCellIntrfIdent.<i>IntrfIdentificationMethod</i>	If the interference source is fixed, set this parameter to UL_MANUAL . Otherwise, set this parameter to UL_AUTO .
UL Interference Auto Detect Thld Offset	NRDUCellIntrfIdent.<i>UlIntrfAutoDetectThldOfs</i>	Use the recommended value.
Interference Band Index	NRDUCellIntrfBand.<i>InterferenceBandIndex</i>	Set this parameter based on the frequency-domain position where interference occurs. A maximum of two interference ranges can be configured.
Start RB	NRDUCellIntrfBand.<i>StartRb</i>	Set the parameters based on the frequency-domain position where interference occurs.
End RB	NRDUCellIntrfBand.<i>EndRb</i>	
Interference Direction	NRDUCellIntrfBand.<i>InterferenceDirection</i>	Set this parameter to UPLINK .

Table 6-3 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
UL Interference Difference Threshold	NRDUCellIntrfIdent.<i>UlIntrfDiffThld</i>	Set this parameter based on the actual situation. A smaller value of this parameter results in a higher probability of triggering uplink interference avoidance. A larger value of this parameter results in a lower probability. If this parameter is set to a value that matches the uplink interference condition, the average cell throughput increases.

Parameter Name	Parameter ID	Setting Notes
UL Interference Freq Sch Judge Offset	NRDUCellIntrfIdent.UlInterFreqSchJudgeOfs	Set this parameter as required. If this parameter is set to an overly large value, scheduling RBs subjected to uplink interference in the case of strong interference decreases UE throughput. If this parameter is set to an overly small value, the frequency segment that was able to be scheduled now cannot be scheduled in the case of weak interference, which reduces the number of RBs to be scheduled and UE throughput.
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommonChnIntrfAvoidSwitch	Select the PUCCH_INTRF_AVOID_SW option when the PUCCH experiences interference.
DL Interference Identification Offset	NRDUCellIntrfIdent.DlInterfIdentOfs	Set this parameter based on the actual situation. A smaller value of this parameter results in a higher probability of triggering downlink interference avoidance. A larger value of this parameter results in a lower probability. The more the match between the parameter value and the actual interference level on the live network, the higher the interference range detection accuracy and downlink UE throughput.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Pay attention to the following during the configuration using MML commands:

- Changing the value of the **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter will cause cells to automatically restart.
- Changing the value of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter will cause cells to automatically restart.

Activation Command Examples

- Enabling external interference avoidance and manual configuration of the uplink interference range
//Enabling manual configuration of the uplink interference range for external interference avoidance and configuring an interference range
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,
IntrfOptimizationMethod=UL_UE_LEVEL_INTRF_AVOID-1, IntrfIdentificationMethod=UL_MANUAL;
ADD NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0, StartRb=0, EndRb=20,
InterferenceDirection=UPLINK;
- Enabling external interference avoidance and automatic identification of the uplink interference range
//Enabling automatic identification of the uplink interference range for external interference avoidance and configuring the interference identification threshold
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,
IntrfOptimizationMethod=UL_UE_LEVEL_INTRF_AVOID-1, IntrfIdentificationMethod=UL_AUTO,
ULIntrfAutoDetectThldOfs=20;

Optimization Command Examples

```
//Configuring the uplink interference threshold difference
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ULIntrfDiffThld=10;
//Configuring an offset used for determining whether an uplink interference range can be scheduled
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ULIntrfFreqSchJudgeOfs=0;
//Enabling PUCCH interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=101, CommChnIntrfAvoidSwitch=PUCCH_INTRF_AVOID_SW-1;
//Configuring the DIntrfIdentOfs parameter
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, DIntrfIdentOfs=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Disabling external interference avoidance and manual configuration of the uplink interference range
RMV NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0;
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, IntrfOptimizationMethod=UL_UE_LEVEL_INTRF_AVOID-0,
IntrfIdentificationMethod=NOT_CONFIG;
- Disabling external interference avoidance and automatic identification of the uplink interference range
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, IntrfOptimizationMethod=UL_UE_LEVEL_INTRF_AVOID-0,
IntrfIdentificationMethod=NOT_CONFIG;

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

- Using the MAE-Access
Enable a single UE with SRS-based weights to perform full buffer services. Observe the number of used downlink RBs in the cell on the MAE-Access. If the number decreases after this function is enabled, this function has taken effect.
- Observing the switch status
Run the **LST NRDUCELLINTRFIDENT** command. If the value of **Interference Identification Method** is **UL_MANUAL** or **UL_AUTO** in the command output, this function has been enabled.

6.2.4.3 Network Monitoring

Observe the average downlink cell throughput, average uplink cell throughput, and average uplink UE throughput to check the increase in the cell rate after this function is enabled.

- The average downlink cell throughput indicated by **Cell Downlink Average Throughput (DU)** is expected to increase.
- The average uplink cell throughput indicated by **Cell Uplink Average Throughput (DU)** is expected to increase.
- The average uplink UE throughput indicated by **User Uplink Average Throughput (DU)** is expected to increase.

6.3 Cell-level PUSCH Interference Avoidance (Low-Frequency TDD)

6.3.1 Principles

Live NR networks can experience interference from various external sources. Typically, external sources of interference affect uplink signals on NR networks. If the PUSCH experiences strong interference, consecutive bit errors occur in the uplink. These bit errors negatively affect network KPIs, such as increasing the service drop rate. An NR cell can avoid scheduling the PUSCH over a specific range of frequencies affected by interference to prevent the interference from affecting the PUSCH if the cell meets both of the conditions: (1) The interference-affected bandwidth is constant, and (2) uplink signals are continuously interfered in the time domain.

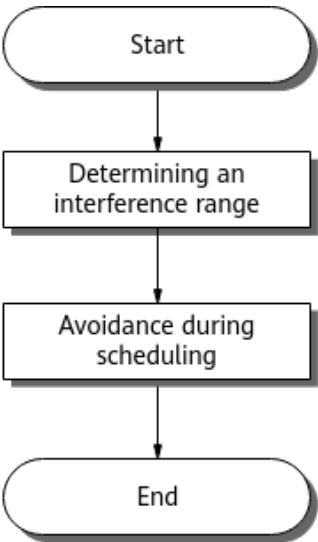
In response, cell-level PUSCH interference avoidance is introduced. **This function achieves interference avoidance during scheduling. The decision to enable this function depends on interference. To evaluate interference on the live network and determine affected frequency-domain positions, O&M personnel can analyze the average interference and noise detected on the PRBs**

available within the entire uplink bandwidth of a cell (for example, indicated by counters N.UL.NI.Avg.PRB0 to N.UL.NI.Avg.PRB272 for a cell with a 100 MHz bandwidth). Alternatively, they can use the cell interference tracing function of the MAE or other interference evaluation tools.

This function can be enabled by applying all the following settings:

- Selecting both the **UL_CELL_LEVEL_INTRF_AVOID** and **DL_UE_LEVEL_INTRF_AVOID** options of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter
- Setting the **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter to **UL_DL_MANUAL**
- Setting the **NRDUCellIntrfBand.InterferenceDirection** parameter to **UPLINK** for at least one interference-affected range or interference-present range (interference range for short)

Figure 6-6 Process of cell-level PUSCH interference avoidance



Following are the details of the process shown in [Figure 6-6](#):

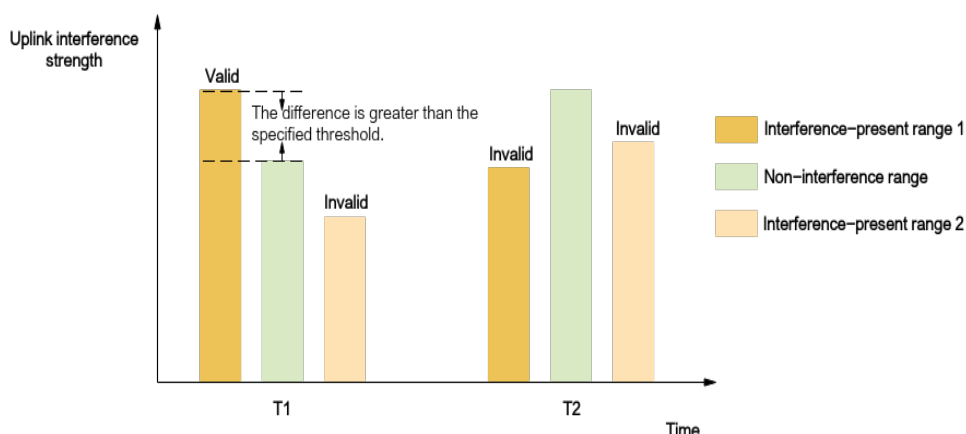
1. Determining an interference range
 - Manual configuration of an uplink interference range

Configuration	Parameter	Description
Selecting manual configuration of an uplink interference range	NRDUCellIntrfIdent.IntrfIdentificationMethod	Set this parameter to UL_DL_MANUAL .
Adding an interference range index	NRDUCellIntrfBand.InterferenceBandIndex	Up to two interference ranges can be configured.

Configuration	Parameter	Description
Configuring an interference range	NRDUCellIntrfBand.StartRb and NRDUCellIntrfBand.EndRb	The parameters are separately used to configure the start RB position and end RB position for each interference range.
Configuring the interference direction for an interference range	NRDUCellIntrfBand.InterferenceDirection	Set this parameter to UPLINK , indicating uplink interference.

- Verifying the validity of the configured interference range
The base station determines whether an uplink interference range is valid using either of the following methods:
 - Method 1 (when optimized uplink interference identification is enabled): The base station compares the difference in the uplink interference strength between the interference and non-interference ranges against the **NRDUCellIntrfIdent.UlIntrfDiffThld** parameter value multiple times within a period of 10 seconds. If the difference is greater than the **NRDUCellIntrfIdent.UlIntrfDiffThld** parameter value for a proportion of times and this proportion is greater than **NRDUCellIntrfIdent.UlIntrfSatisfactionThld** parameter value, the interference range is considered to be valid; otherwise, the interference range is considered to be invalid. This method fits a scenario in which interference fluctuates greatly in time domain.
 - Method 2 (when optimized uplink interference identification is disabled): The base station compares the difference in the uplink interference strength between the interference and non-interference ranges against the **NRDUCellIntrfIdent.UlIntrfDiffThld** parameter value at an interval of 100 ms. If the difference is greater than this parameter value, the interference range is considered to be valid; otherwise, the interference range is considered to be invalid. For further clarification, see [Figure 6-7](#). This method fits a scenario where interference is stable in the time domain.

Figure 6-7 Verifying the validity of an interference range



The optimized uplink interference identification function is controlled by the **UL_INTRF_IDENT_OPT_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter.

2. Avoidance during scheduling
 - Uplink scheduling:
 - When an uplink interference range is valid, uplink scheduling avoids this range, which prevents interference from affecting the PUSCH but reduces the resources available for PUSCH scheduling.
 - When an uplink interference range is invalid, uplink scheduling can use resources within this range.
 - Downlink scheduling:
 - Scheduling for UEs that rely on SRS-based weights: If an uplink interference range is manually configured and the difference in the uplink interference strength between the interference and non-interference ranges is greater than the sum of the **NRDUCellIntrfIdent.UlIntrfDiffThld** and **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter values, downlink scheduling avoids this range; otherwise, downlink scheduling can use the resources within this range.

NOTE

If interference avoidance is needed only for uplink scheduling (not for PDSCH scheduling), it is recommended to set the **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter to **100**. If interference avoidance is needed for PDSCH scheduling, set this parameter according to specific network requirements.

- Scheduling for UEs that rely on PMI-based weights: An interference range can be used for downlink SU-MIMO scheduling and beam-domain pairing for downlink MU-MIMO, but not for correlation-based UE pairing for downlink MU-MIMO.

For details about SRS- and PMI-based weights, see "Downlink Beamforming" in *MIMO (TDD)*.

6.3.2 Network Analysis

6.3.2.1 Benefits

This function applies to a cell where the uplink frequency band is partially interfered. It can be enabled for a cell to lessen the impact of interference on the service drop rate if a cell meets both of the conditions: (1) The strength of interference affecting the specified uplink range is more than 10 dB higher than the average interference strength of the entire band, and (2) the service drop rate is adversely affected. Even after this function is enabled, the service drop rate may not necessarily decrease. This is because many reasons cause service drops and interference is only one of them.

6.3.2.2 Impacts

The impacts between this function and other functions are described in [6.3.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

- Interference avoidance in a cell reduces the bandwidths available for uplink and downlink data channels. The reduction depends on the specified interference-affected bandwidth.
- If the difference in uplink interference strength between the interference and non-interference ranges is greater than the sum of the **NRDUCellIntrfIdent.UlIntrfDiffThld** and **NRDUCellIntrfIdent.DlIntrfIdentOfs** parameter values, the measurement of the average interference on SRS excludes data from the interference range. As a result, the **N.SRS.NI.AVG** value decreases.

6.3.3 Requirements

6.3.3.1 Licenses

None

6.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.3.2.1 Prerequisite Functions

None

6.3.3.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
External interference avoidance (low-frequency TDD)	UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimization-Method parameter, and NRDUCellIntrfIdent.IntrfIdentification-Method set to UL_MANUAL or UL_AUTO	<i>Interference Avoidance</i>	None
VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	None
3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	The two functions cannot be both enabled: cell-level PUSCH interference avoidance and 3GPP R15-compliant uplink slot aggregation.
3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	The two functions cannot be both enabled: cell-level PUSCH interference avoidance and 3GPP R16-compliant uplink slot aggregation.
Distributed Massive MIMO Service Switch	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	Cell-level PUSCH interference avoidance cannot be enabled if the DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter is selected.

Function Name	Function Switch	Reference	Description
SRS Power Difference Rank Select Switch	SRS_DIFF_BASED_RANK_SELECT_SW option of the NRDUCellAlgoSwitch . <i>RankSelectAlgoSw</i> parameter	None	Cell-level PUSCH interference avoidance cannot be enabled if the SRS_DIFF_BASED_RANK_SELECT_SW option of the NRDUCellAlgoSwitch . <i>RankSelectAlgoSw</i> parameter is selected.
PUCCH interference avoidance	PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch . <i>CommChnIntrfAvoidSwitch</i> parameter	<i>Interference Avoidance</i>	The two functions cannot be both enabled: cell-level PUSCH interference avoidance and PUCCH interference avoidance.
High-Resolution 2T Weight Switch	HR_2T_WEIGHT_SW option of the NRDUCellAlgoSwitch . <i>BeamOptSwitch</i> parameter	None	If the HR_2T_WEIGHT_SW option of the NRDUCellAlgoSwitch . <i>BeamOptSwitch</i> parameter is selected, cell-level PUSCH interference avoidance cannot be enabled.
CORESET 0 offset policy configuration	NRDUCellPdcchAlgo . <i>Coreset0OffsetPolicy</i> set to EDGE_FIRST	<i>Channel Management</i>	The two functions cannot be both enabled: cell-level PUSCH interference avoidance and CORESET 0 offset policy configuration.

6.3.3.2.3 Function Impacts

None

6.3.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules that work in bands n41, n77, n78, and n79 support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.3.4 Cells

The cell bandwidth must be greater than or equal to 30 MHz.

6.3.3.5 Others

None

6.3.4 Operation and Maintenance

6.3.4.1 Data Configuration

6.3.4.1.1 Data Preparation

Table 6-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Interference Optimization Method	NRDUCellIntrfIdent.IntrfOptimizationMethod	Select the UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options and deselect other options of this parameter.
Interference Identification Method	NRDUCellIntrfIdent.IntrfIdentificationMethod	Set this parameter to UL_DL_MANUAL .

Parameter Name	Parameter ID	Setting Notes
Interference Band Index	NRDUCellIntrfBand.InterferenceBandIndex	- For a cell with the NRDUCellIntrfBand.InterferenceDirection parameter set to UPLINK only, you can configure a maximum of two interference ranges. - For a cell with the NRDUCellIntrfBand.InterferenceDirection parameter set to both UPLINK_DOWNLINK and UPLINK, you can configure a maximum of three interference ranges. Out of these, a maximum of two can be configured for UPLINK_DOWNLINK and one for UPLINK.
Start RB	NRDUCellIntrfBand.StartRb	Set the parameters based on the frequency-domain position where interference occurs.
End RB	NRDUCellIntrfBand.EndRb	
Interference Direction	NRDUCellIntrfBand.InterferenceDirection	Set this parameter to UPLINK .
UL Interference Difference Threshold	NRDUCellIntrfIdent.UlIntrfDiffThld	Set this option as required. A smaller value of this parameter results in a higher probability of triggering uplink interference avoidance. A larger value of this parameter results in a lower probability. The more precisely this parameter is tuned to match the uplink interference conditions, the higher the average cell throughput will be.
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	It is recommended that the UL_INTRF_IDENT_OPT_SW option be selected if interference fluctuates greatly in the time domain.

Parameter Name	Parameter ID	Setting Notes
Uplink Interference Satisfaction Threshold	NRDUCellIntrfIdent.UlIntrfSatisfactionThld	Set this parameter as required if the UL_INTRF_IDENT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter is selected. A smaller value of this parameter results in a higher probability that uplink interference avoidance takes effect. A larger value of this parameter results in a lower probability that uplink interference avoidance takes effect and a lower probability that the uplink user-perceived rate increases. The more precisely this parameter is tuned to match the uplink interference conditions, the higher the average cell throughput will be.
DL Interference Identification Offset	NRDUCellIntrfIdent.DlIntrfIdentOfs	If interference avoidance is not needed for PDSCH scheduling, set NRDUCellIntrfIdent.DlIntrfIdentOfs to 100.

6.3.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

Points to consider when using MML commands:

- A cell automatically restarts after the value of the **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter is changed.
- A cell automatically restarts after the value of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter is changed.

Activation Command Examples

```
//Enabling cell-level PUSCH interference avoidance and configuring an interference range
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, IntrfIdentificationMethod=UL_DL_MANUAL,
IntrfOptimizationMethod=UL_CELL_LEVEL_INTRF_AVOID-1&DL_UE_LEVEL_INTRF_AVOID-1,
ULIntrfDiffThld=15;
ADD NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0, StartRb=164, EndRb=219,
InterferenceDirection=UPLINK;
//(Optional) Enabling optimized uplink interference identification and configuring Uplink Interference
Satisfaction Threshold. This step is recommended in scenarios where interference fluctuates greatly in the
time domain.
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=UL_INTRF_IDENT_OPT_SW-1,
ULIntrfSatisfactionThld=30;
//Setting DL Interference Identification Offset to 100 if interference avoidance is not needed for PDSCH
scheduling
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, DLIntrfIdentOfs=100;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing an uplink interference range and disabling cell-level PUSCH interference avoidance
RMV NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0;
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,
IntrfOptimizationMethod=UL_CELL_LEVEL_INTRF_AVOID-0&DL_UE_LEVEL_INTRF_AVOID-0,
IntrfIdentificationMethod=NOT_CONFIG;
```

6.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.3.4.2 Activation Verification

Using the MAE-Access: In a full-buffer scenario, check the number of used uplink RBs. If the number decreases compared to that obtained before this function is enabled, this function has taken effect.

6.3.4.3 Network Monitoring

After this function is enabled, observe the following KPIs to determine how the uplink cell throughput and service drop rate have changed:

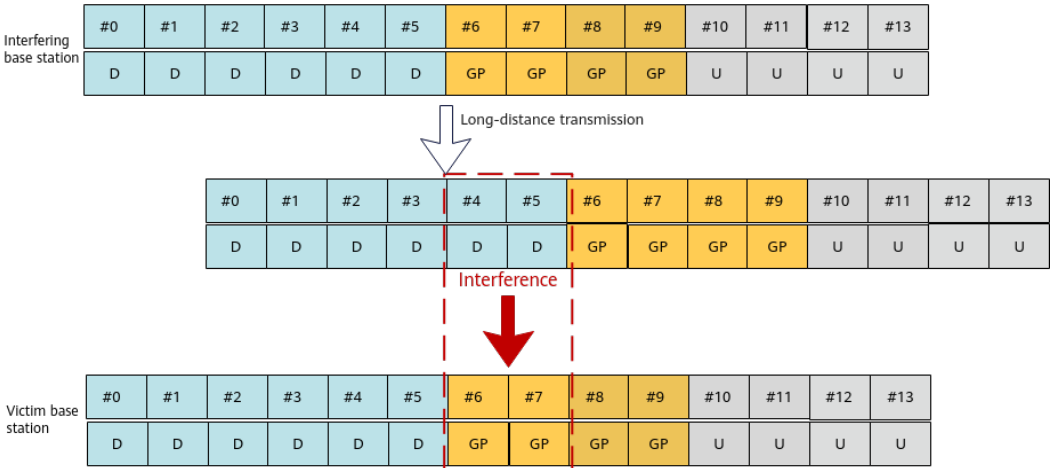
- **Cell Uplink Average Throughput (DU)**, which is expected to decrease
- **Service Call Drop Rate (CU)** and **Service Call Drop Rate (CU, Inactive)**, which are expected to decrease

6.4 GP Symbol Shutdown in Special Slots

6.4.1 Principles

Due to long-distance transmission of downlink signals, a victim base station may receive downlink signals from other base stations in GP symbols, as shown in [Figure 6-8](#). If these interfering downlink signals are strong, the RF module's interference protection mechanism is triggered in the corresponding GP symbols, affecting the reception of subsequent uplink symbols. As a result, uplink user experience may deteriorate. For a wideband RF module, GP symbols are more susceptible to interference due to wideband uplink reception. To address this issue, the GP symbol shutdown in special slots function is introduced.

Figure 6-8 Interference scenario applicable to GP symbol shutdown in special slots



After GP symbol shutdown in special slots is enabled, if no data is transmitted in the first two GP symbols of a special slot, the gNodeB shuts down these GP symbols and postpones the timing of downlink-to-uplink switching, thereby reducing the impact of interference in the GP symbols on uplink performance. This function is controlled by the **SS_GAP_SYMBOL_SHUTDN_SW** option of the **NRDUCellTimeIntrf. TimeIntrfAvoidSw** parameter.

In LNR multi-mode scenarios (where the RF module also serves an LTE cell), the GP symbol shutdown in special subframes function must also be enabled for the LTE cell. For details, see section "GP Symbol Shutdown in Special Subframes" of *Interference Detection and Suppression* in eRAN feature documentation.

6.4.2 Network Analysis

6.4.2.1 Benefits

When the RF module's interference protection mechanism is triggered due to strong interference in GP symbols, enabling this function mitigates the impact of

uplink interference on uplink user experience, decreases the value of **N.GAP.ThirdSymbolFromLast.Pwr**, and increases the **User Uplink Average Throughput (DU)**.

6.4.2.2 Impacts

The impacts between this function and other functions are described in [6.4.3.2.3 Function Impacts](#).

There are no impacts on the network.

6.4.3 Requirements

6.4.3.1 Licenses

None

6.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.4.3.2.1 Prerequisite Functions

None

6.4.3.2.2 Mutually Exclusive Functions

None

6.4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
8_2_SS54 adaptation	8_2_SS54_ADAPT_SW option of the gNodeBAlgo.UISchSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCell.SlotStructure parameter is set to SS54 and both GP symbol shutdown in special slots and 8_2_SS54 adaptation are enabled, 8_2_SS54 adaptation takes effect preferentially.

Function Name	Function Switch	Reference	Description
Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both uplink symbol power saving and GP symbol shutdown in special slots are enabled, the value of VS.RADIOEQM.UL.SymbolShutdown.Dur increases due to GP symbol shutdown.

6.4.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only the AAU5685md supports this function.

6.4.3.4 Cells

- The **NRDUCell.SlotAssignment** parameter is set to **8_2_DDDDDDDDSUU** or **8_2_DDDSUUDDDD**.
- The **NRDUCell.SlotStructure** parameter is set to **SS54** or **SS84**.

6.4.3.5 Others

None

6.4.4 Operation and Maintenance

6.4.4.1 Data Configuration

6.4.4.1.1 Data Preparation

Table 6-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Time Interference Avoidance Switch	NRDUCellTimeIntrf.AvoidSw	SS_GAP_SYMBOL_SHUTDOWN_SW	It is recommended that this option be selected.

6.4.4.1.2 Using MML Commands

Before using MML commands, refer to [6.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling GP symbol shutdown in special slots
MOD NRDUCELLTIMEINTRF: NrDuCellId=101, TimeIntrfAvoidSw=SS_GAP_SYMBOL_SHUTDOWN_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling GP symbol shutdown in special slots
MOD NRDUCELLTIMEINTRF: NrDuCellId=101, TimeIntrfAvoidSw=SS_GAP_SYMBOL_SHUTDOWN_SW-0;
```

6.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.4.2 Activation Verification

- Run the **LST NRDUCELLTIMEINTRF** command. If **Time Interference Avoidance Switch** has a value of **Special Slot GP Symbol Shutdown Switch:On**, GP symbol shutdown in special slots has been enabled.
- After this function is enabled, observe the decrease in the value of the **N.GAP.ThirdSymbolFromLast.Pwr** counter to check whether this function has

taken effect. If the value of **N.GAP.ThirdSymbolFromLast.Pwr** decreases, this function has taken effect.

6.4.4.3 Network Monitoring

After this function is enabled, monitor the following indicators:

- **N.GAP.ThirdSymbolFromLast.Pwr**
- **User Uplink Average Throughput (DU)**

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Downlink Scheduling*
- *Cell Management*
- *CoMP*
- *Channel Management*
- *3900 & 5900 Series Base Station Product Documentation*
- *UE Power Saving*
- *Hyper Cell*
- *Cell Combination*
- *URLLC*
- *MIMO (TDD)*
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Indoor Coverage*
- *Energy Conservation and Emission Reduction*
- *3D Networking Experience Improvement*
- *Co-MM (Low-Frequency TDD)*
- **(Video) Common Channel Interference Avoidance**
- **(Video) External Interference Avoidance**
- *Coordinated Interference Management (Low-Frequency TDD)*
- *RedCap*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *VoNR*
- *License Control Item Lists*
- *Co-MM (Low-Frequency TDD)*
- *Remote Interference Management (Low-Frequency TDD)*
- *Fusion Cell (Low-Frequency TDD)*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *Standards Compliance*

- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*